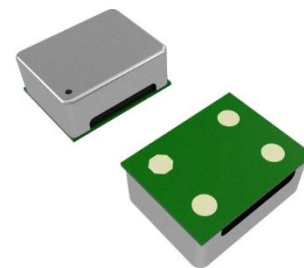


### Description

The AOC97 series is an Oven Controlled Crystal Oscillator (OCXO) offered in a 9.7mm x 7.5mm x 3.9mm four-pad SMD package. Tight frequency stability of  $\pm 10$ ppb over an extended operating temperature range of  $-40^{\circ}\text{C}$  to  $+95^{\circ}\text{C}$  is achieved using an SC-Cut, High "Q" resonator-based design. This series offers a CMOS-compatible output with a 3.3Vdc  $\pm 5\%$  supply voltage, common for most communication infrastructure, base station, and test and measurement equipment applications. The AOC97 series offers industry standard frequencies in the range of 10MHz to 48MHz with low long-term aging and excellent phase noise.



### Features

- SC-Cut, High "Q" resonator-based design
- 3.3Vdc supply voltage
- CMOS compatible output logic
- 9.7mm x 7.5mm x 3.9mm four pad SMD package
- Stability over temperature:  $\pm 10$ ppb over  $-40^{\circ}\text{C}$  to  $+95^{\circ}\text{C}$
- Low long-term aging:  $\pm 500$ ppb over first year
- Industry standard frequencies available
- Excellent phase noise

### Typical Applications

- Cellular infrastructure; Base stations
- Test & measurement equipment
- Switches & routers
- Time & frequency references
- Precision GPS
- Satellite Timing and Frequency
- High End Synthesizers
- Oil and Gas Exploration

### Electrical Specifications [Note 1]

| Parameters                                                                   | Min.                           | Typ. | Max.      | Units              | Notes                                        |
|------------------------------------------------------------------------------|--------------------------------|------|-----------|--------------------|----------------------------------------------|
| Frequency Range                                                              | 10                             |      | 48        | MHz                |                                              |
| Standard Available Frequencies                                               | 10, 19.2, 20, 30.72, 38.88, 48 |      |           | MHz                | Contact Abracon for non-standard frequencies |
| Supply Voltage (Vdd)                                                         | 3.135                          | 3.3  | 3.456     | V                  |                                              |
| Input Power (warm-up)                                                        |                                |      | 2.1       | W                  |                                              |
| Input Power (steady-state)                                                   |                                |      | 0.8       | W                  |                                              |
| Operating Temperature Range                                                  | $-40$                          |      | $+95$     | $^{\circ}\text{C}$ |                                              |
| Storage Temperature Range                                                    | $-55$                          |      | $+105$    | $^{\circ}\text{C}$ |                                              |
| Initial Frequency Tolerance <small>[Note 2]</small>                          |                                |      | $\pm 1$   | ppm                |                                              |
| Frequency Stability over Operating Temperature Range <small>[Note 3]</small> |                                |      | $\pm 10$  | ppb                |                                              |
| Temperature Slope <small>[Note 4]</small>                                    |                                |      | $\pm 1$   | ppb                |                                              |
| Stability vs. Supply Voltage                                                 |                                |      | $\pm 5$   | ppb                | Vdd varied from 3.135V to 3.465V             |
| Stability vs. Load                                                           |                                |      | $\pm 5$   | ppb                | 5% load change                               |
| Aging per Day                                                                |                                |      | $\pm 3$   | ppb                | after 30 days of operation                   |
| Aging per Year                                                               |                                |      | $\pm 500$ | ppb                | after 30 days of operation                   |
| Warm-up Time <small>[Note 5]</small>                                         |                                |      | 3         | minutes            |                                              |
| Start-up Time <small>[Note 6]</small>                                        |                                |      | 50        | ms                 |                                              |
| <b>Output Waveform</b>                                                       |                                |      |           |                    |                                              |
| High-level Output Voltage (V <sub>OH</sub> )                                 | 2.4                            |      |           | V                  |                                              |
| Low-level Output Voltage (V <sub>OL</sub> )                                  |                                |      | 0.4       | V                  |                                              |
| Output Signal                                                                |                                | CMOS |           |                    |                                              |
| Output Load                                                                  |                                | 15   |           | pF                 |                                              |
| Rise and Fall Time (t <sub>r</sub> , t <sub>f</sub> )                        |                                |      | 6         | ns                 | 10% to 90% of waveform                       |
| Duty Cycle                                                                   | 45                             |      | 55        | %                  | @50% of waveform                             |

**Electrical Specifications** *continued* [Note 1]

| Parameters                 | Min. | Typ. | Max. | Units  | Notes         |
|----------------------------|------|------|------|--------|---------------|
| Phase Noise (@ 10.0000MHz) |      | -80  | -70  | dBc/Hz | 1Hz offset    |
|                            |      | -115 | -110 |        | 10Hz offset   |
|                            |      | -146 | -138 |        | 100Hz offset  |
|                            |      | -157 | -151 |        | 1kHz offset   |
|                            |      | -161 | -156 |        | 10kHz offset  |
|                            |      | -164 | -159 |        | 100kHz offset |
|                            |      | -165 | -161 |        | 1MHz offset   |
| Phase Noise (@ 19.2000MHz) |      | -80  | -70  | dBc/Hz | 1Hz offset    |
|                            |      | -115 | -110 |        | 10Hz offset   |
|                            |      | -146 | -138 |        | 100Hz offset  |
|                            |      | -160 | -154 |        | 1kHz offset   |
|                            |      | -164 | -159 |        | 10kHz offset  |
|                            |      | -164 | -160 |        | 100kHz offset |
|                            |      | -165 | -161 |        | 1MHz offset   |
| Phase Noise (@ 20.0000MHz) |      | -75  | -65  | dBc/Hz | 1Hz offset    |
|                            |      | -113 | -108 |        | 10Hz offset   |
|                            |      | -146 | -138 |        | 100Hz offset  |
|                            |      | -160 | -154 |        | 1kHz offset   |
|                            |      | -163 | -158 |        | 10kHz offset  |
|                            |      | -163 | -158 |        | 100kHz offset |
|                            |      | -163 | -158 |        | 1MHz offset   |
| Phase Noise (@30.7200MHz)  |      | -73  | -63  | dBc/Hz | 1Hz offset    |
|                            |      | -110 | -103 |        | 10Hz offset   |
|                            |      | -140 | -130 |        | 100Hz offset  |
|                            |      | -158 | -152 |        | 1kHz offset   |
|                            |      | -165 | -160 |        | 10kHz offset  |
|                            |      | -165 | -160 |        | 100kHz offset |
|                            |      | -165 | -160 |        | 1MHz offset   |
| Phase Noise (@ 38.8800MHz) |      | -67  | -57  | dBc/Hz | 1Hz offset    |
|                            |      | -105 | -97  |        | 10Hz offset   |
|                            |      | -135 | -125 |        | 100Hz offset  |
|                            |      | -156 | -151 |        | 1kHz offset   |
|                            |      | -165 | -160 |        | 10kHz offset  |
|                            |      | -165 | -160 |        | 100kHz offset |
|                            |      | -165 | -160 |        | 1MHz offset   |
| Phase Noise (@ 48.0000MHz) |      | -60  | -50  | dBc/Hz | 1Hz offset    |
|                            |      | -99  | -90  |        | 10Hz offset   |
|                            |      | -130 | -120 |        | 100Hz offset  |
|                            |      | -155 | -150 |        | 1kHz offset   |
|                            |      | -165 | -160 |        | 10kHz offset  |
|                            |      | -165 | -160 |        | 100kHz offset |
|                            |      | -165 | -160 |        | 1MHz offset   |

Note 1: All measurements guaranteed at +25°C, Vdd = 3.3V, CL = 15pf unless otherwise specified.

Note 2: Measured at +25°C, Vdd=3.3V within 30 days, at time of shipment.

Note 3: Varied from -40 to +95°C, @ fref =(fmax +fmin )/2, Vdd=3.3V, CL=15pf, less than 2°C per minute

Note 4: Temperature slope ±1°C/min with any temperature window over -40 to +95°C

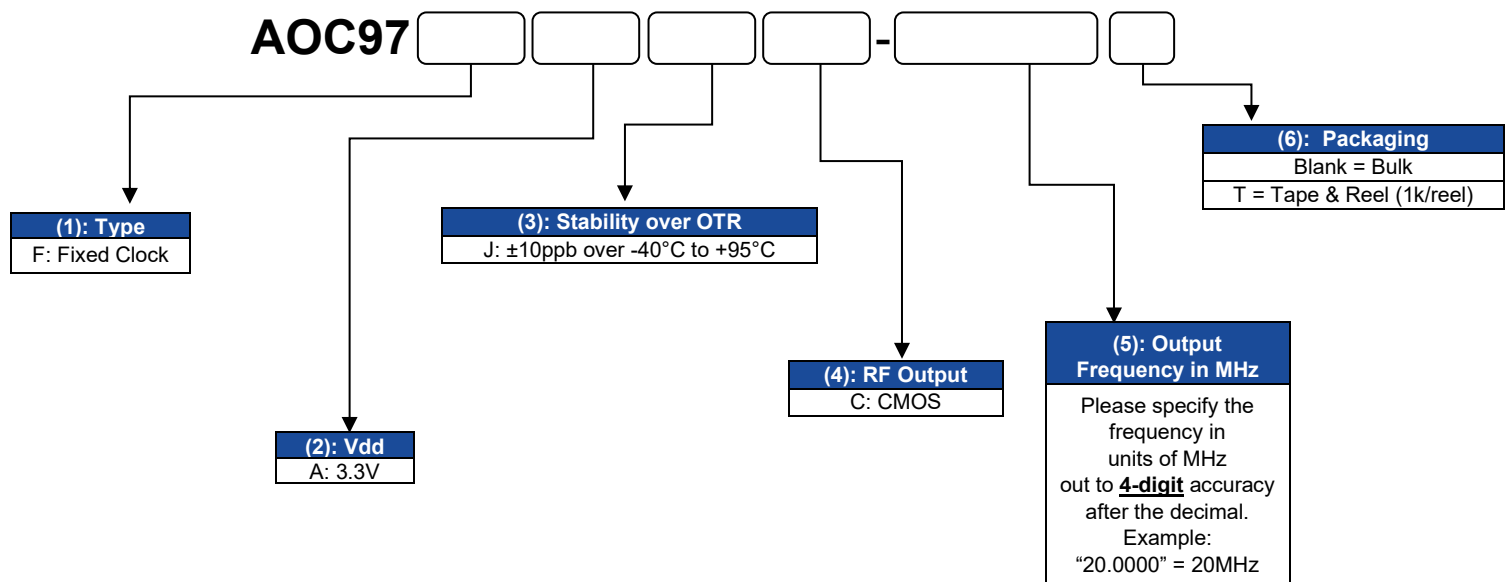
Note 5: Time until RF output is within ± 0.1 ppm referenced to last frequency reading 1 hour after startup TA=25°C.

Note 6: Time until RF output waveform is within output logic levels, duty cycle and rise/fall time specifications.

## Environmental and Mechanical

| Parameters    | Description |
|---------------|-------------|
| MSL           | 3           |
| REACH/RoHS II | Compliant   |
| ESD           | Sensitive   |

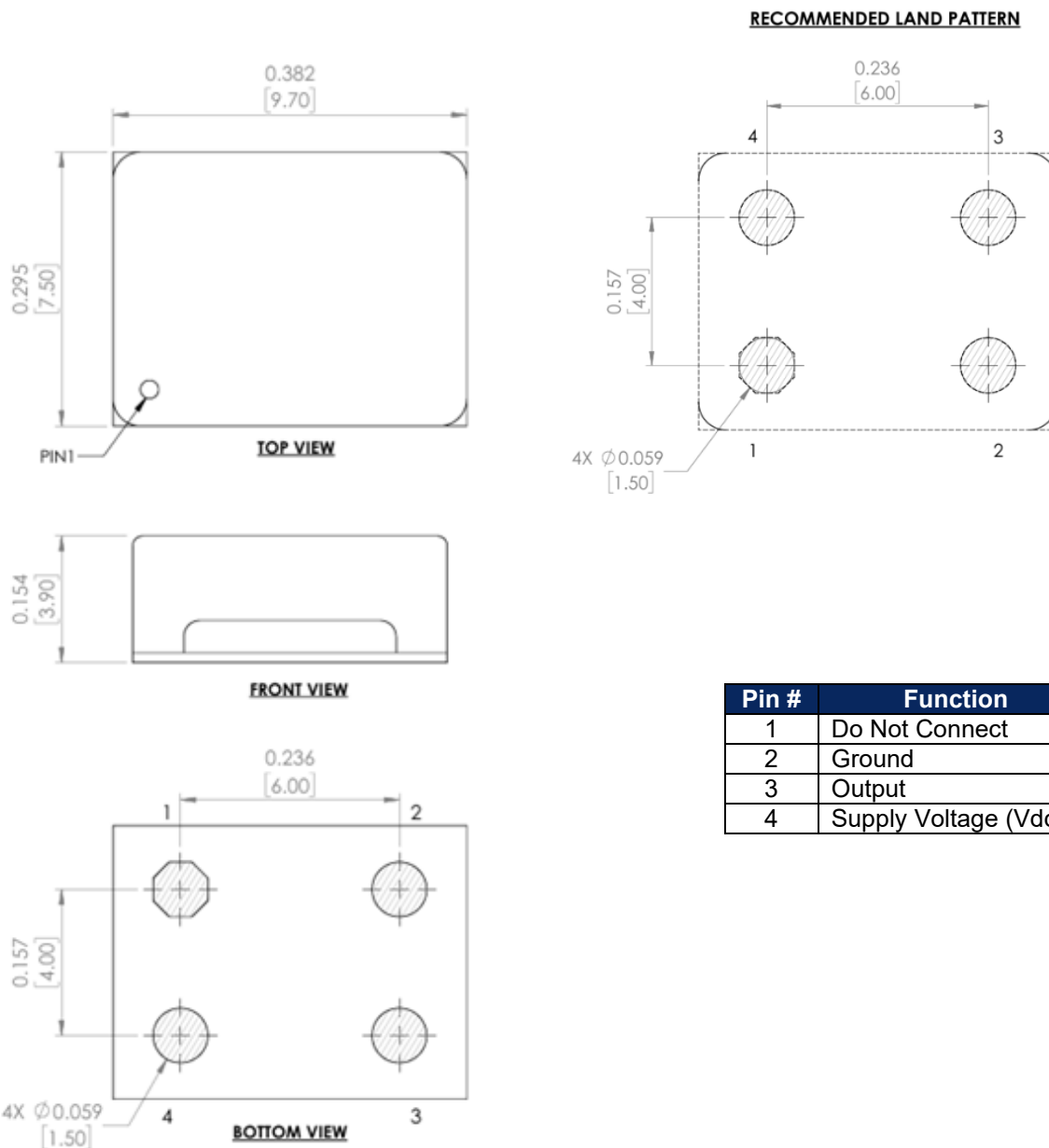
## Part Identification



## Part Number Example:

**AOC97FAJC-20.0000T**

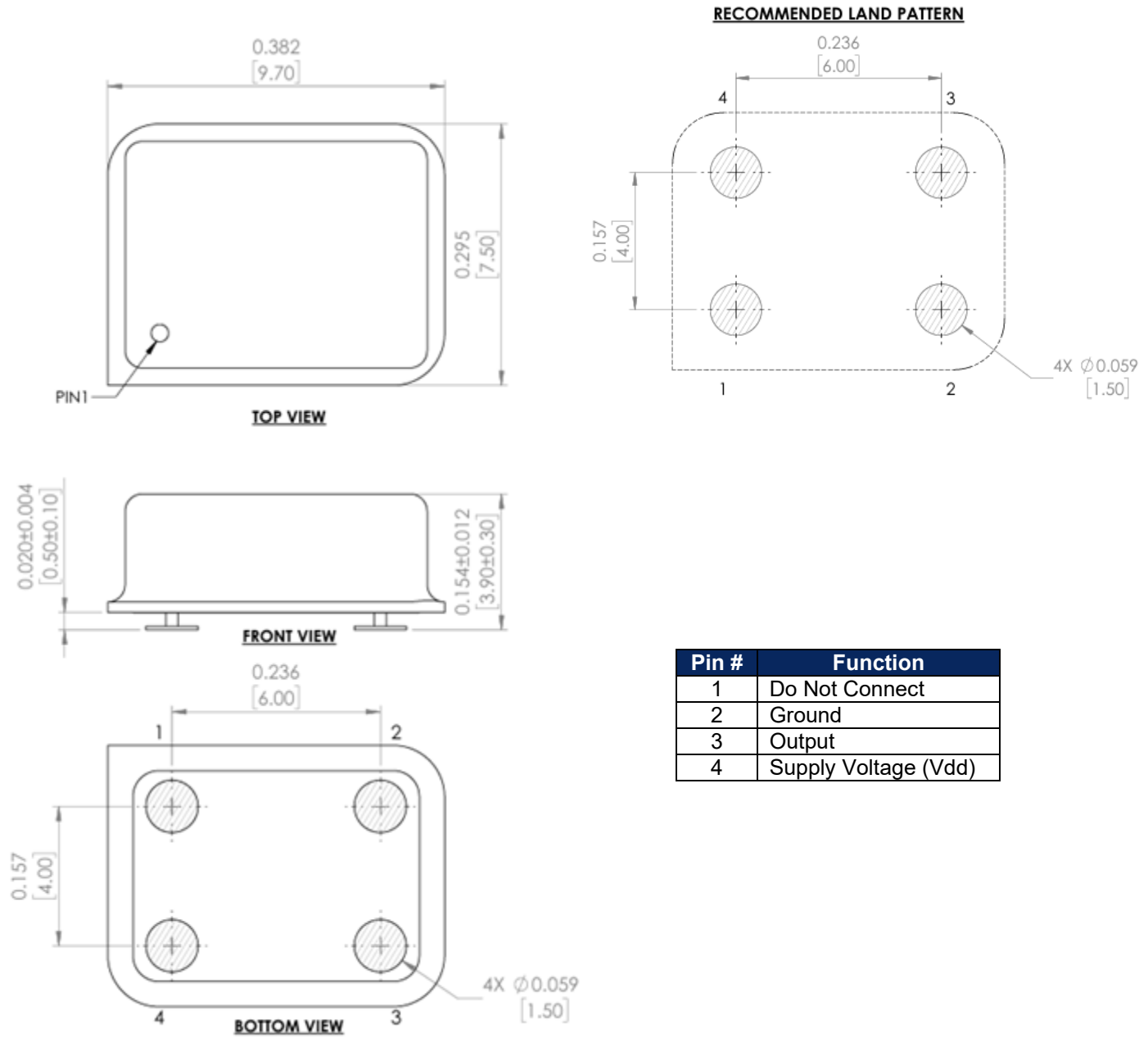
**Mechanical Dimensions (10 MHz, 19.2 MHz, 30.72 MHz, 38.88 MHz)**



**Dimensions: inches [mm]**

**Tolerance  $\pm 0.3\text{mm}$  without mark**

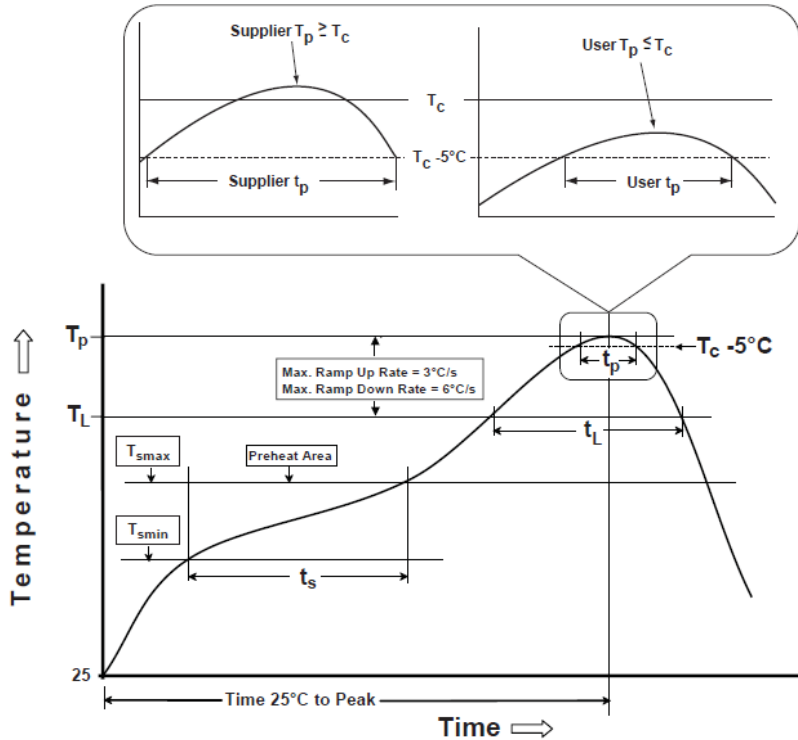
**Mechanical Dimensions (20 MHz, 48 MHz)**



**Dimensions: inches [mm]**

**Tolerance  $\pm 0.3\text{mm}$  without mark**

**Reflow Profile [JEDEC J-STD-020]**



**Table 1**

| SnPb Eutectic Process<br>Classification Temperatures ( $T_c$ ) |                             |                             |
|----------------------------------------------------------------|-----------------------------|-----------------------------|
| Package Thickness                                              | Volume mm <sup>3</sup> <350 | Volume mm <sup>3</sup> ≥350 |
| <2.5 mm                                                        | 235 °C                      | 220 °C                      |
| ≥2.5 mm                                                        | 220 °C                      | 220 °C                      |

**Table 2**

| Pb-Free Process<br>Classification Temperatures ( $T_c$ ) |                             |                                 |                              |
|----------------------------------------------------------|-----------------------------|---------------------------------|------------------------------|
| Package Thickness                                        | Volume mm <sup>3</sup> <350 | Volume mm <sup>3</sup> 350-2000 | Volume mm <sup>3</sup> >2000 |
| <1.6 mm                                                  | 260 °C                      | 260 °C                          | 260 °C                       |
| 1.6 mm - 2.5 mm                                          | 260 °C                      | 250 °C                          | 245 °C                       |
| >2.5 mm                                                  | 250 °C                      | 245 °C                          | 245 °C                       |

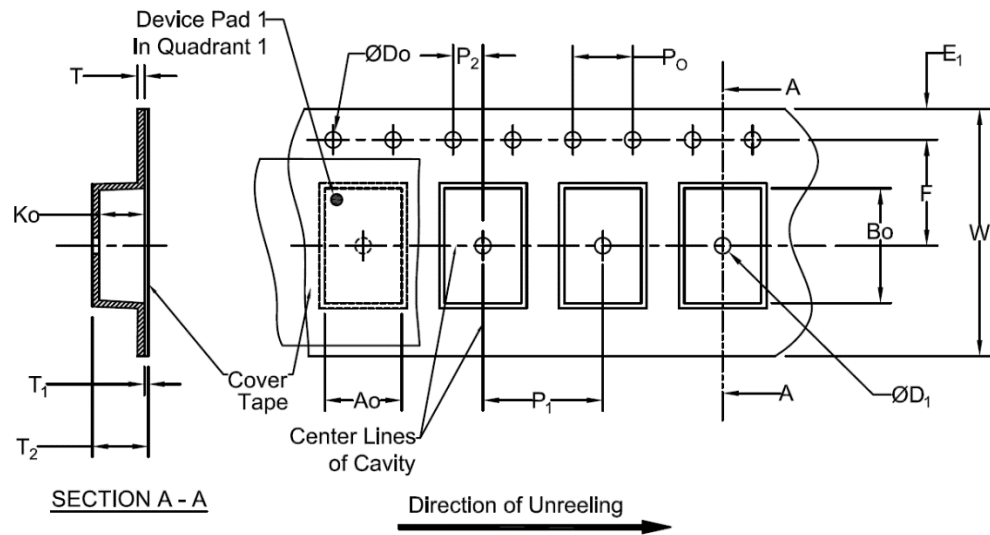
| Profile Feature                                                                   | Sn-Pb Eutectic Assembly | Pb-Free Assembly   |
|-----------------------------------------------------------------------------------|-------------------------|--------------------|
| Preheat / soak                                                                    |                         |                    |
| Temperature minimum ( $T_{smin}$ )                                                | 100°C                   | 150°C              |
| Temperature maximum ( $T_{smax}$ )                                                | 150°C                   | 200°C              |
| Time ( $T_{smin}$ to $T_{smax}$ ) ( $t_s$ )                                       | 60 - 120 sec.           | 60 - 120 sec.      |
| Average ramp-up rate ( $T_{smax}$ to $T_p$ )                                      | 3°C/sec. max            | 3°C/sec. max       |
| Liquidous temperature ( $T_L$ )                                                   | 183°C                   | 217°C              |
| Time at liquidous ( $t_L$ )                                                       | 60 - 150 sec.           | 60 - 150 sec.      |
| Peak package body temperature ( $T_p$ )*                                          | <b>see Table 1</b>      | <b>see Table 2</b> |
| Time ( $t_p$ )** within 5°C of the specified classification temperature ( $T_c$ ) | 20 sec.                 | 30 sec.            |
| Ramp-down rate ( $T_p$ to $T_{smax}$ )                                            | 6°C/sec. max            | 6°C/sec. max       |
| Time 25°C to peak temperature                                                     | 6 min. max              | 8 min. max         |
| Reflow cycles                                                                     | 2 max                   | 2 max              |

\*Tolerance for peak profile temperature ( $T_p$ ) is defined as a supplier minimum and a user maximum.

\*\*Tolerance for time at peak profile temperature ( $t_p$ ) is defined as supplier minimum and a user maximum.

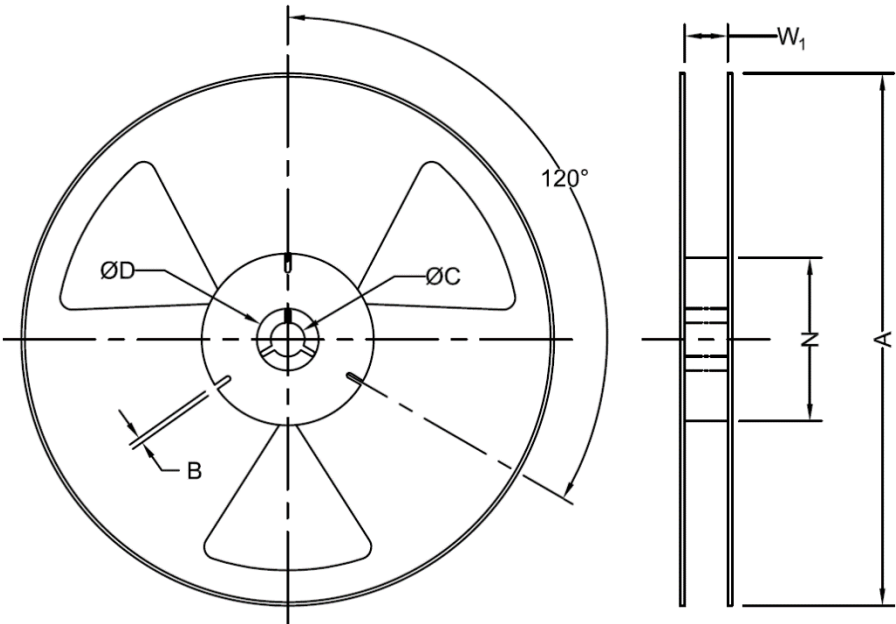
Packaging

T = 1000 pcs/reel



| Tape Specifications (mm) |                |                |                |                      |                      |                      |         |
|--------------------------|----------------|----------------|----------------|----------------------|----------------------|----------------------|---------|
| Width                    | Ao             | Bo             | Do             | D <sub>1</sub> (Min) | E <sub>1</sub>       | F                    | Ko      |
| 16mm                     | *              | *              | 1.5+0.1/-0.0   | 1.50                 | 1.75±0.1             | 7.125                | *       |
| Width                    | P <sub>1</sub> | P <sub>2</sub> | P <sub>0</sub> | T (Max)              | T <sub>1</sub> (Max) | T <sub>2</sub> (Max) | W (Max) |
| 16mm                     | 12.0±0.1       | 2.0±0.1        | 4.0±0.1        | 0.3                  | 0.1                  | 8.0                  | 16.0    |

\*Note: Compliant to EIA-481



| Reel Specifications (mm) |          |         |         |          |        |          |                 |
|--------------------------|----------|---------|---------|----------|--------|----------|-----------------|
| Width                    | Qty/Reel | A       | B       | C        | D      | N        | *W <sub>1</sub> |
| 16mm                     | 1000     | 330±1.0 | 2.5±0.3 | 13.0±0.2 | 22±0.6 | 99.5±0.5 | 16.8+1.0/-0.2   |

\*Note: Measured at Hub